

AKANKSHA MATE

📍 Chicago, IL 📞 +1-312-478-6689 ✉️ amate1@hawk.iit.edu 🔗 <https://www.linkedin.com/in/akanksha-mate-4848a01b4/>

SUMMARY

Accomplished Data Analyst with hands-on experience in data analysis, visualization, and machine learning, skilled in leveraging Python, SQL, and Power BI to drive actionable insights. Optimized data workflows, reducing processing time by **30%**, and improved reporting systems, leading to a **25%** increase in decision-making efficiency. Developed interactive dashboards that supported over **15 key business stakeholders**, driving a **20% boost in ROI** through actionable insights. Seeking a role to apply my technical expertise and problem-solving skills to drive data-driven growth in a dynamic environment.

EDUCATION

Illinois Institute of Technology Jan 2024 - Dec 2025

MAS, Information Technology and Management GPA: 3.66

- **Coursework:** Advanced topics in data management, Programming for data analytics, big data technologies

Sandip Institute of Technology and Research Centre Jul 2019 - May 2023

Bachelor of Computer Engineering GPA: 3.5

- **Coursework:** Database Management System, Data science and big data analytics, Machine learning, Deep learning

WORK EXPERIENCE

Data Analyst Intern | Sahajnanda Tech Services Mar 2023 - Nov 2023

- Automated ETL processes to collect, clean, and manage over **500,000 monthly data points** from databases and APIs, enhancing accuracy and reliability through robust data validation and error handling using Python.
- Transformed raw data into analysis-ready formats and conducted EDA with Tableau and Python, identifying trends and applying statistical analysis to support data-driven decision-making.
- Built and maintained **5+ quarterly dashboards** in Tableau and Excel, providing actionable insights and recommendations that supported cross-departmental strategic initiatives.
- Established comprehensive data validation processes and automated checks, reducing manual effort by **15%** and effectively resolving potential data errors.
- Developed interactive dashboards and visualizations that increased user engagement by **20%** and supported strategic decision-making with clear data-driven reports.

Data Science and Machine Learning Intern | Twowaits Technologies Mar 2022 - Apr 2022

- Developed and implemented machine learning algorithms using Python (Pandas, Scikit-Learn), enhancing model predictive performance by **10%** and aligning with robust data validation techniques.
- Collaborated with cross-functional teams to analyze internal data and identify trends, leading data preprocessing efforts that cleaned, standardized, and validated **10,000 records** to improve overall dataset quality.

PROJECTS

Income Inequality Analysis | R, ggplot2, Hypothesis Testing, Regression, Statistical Modeling Nov 2024 - Dec 2024

- Analyzed gender-based income disparities using hypothesis testing, regression modeling, and correlation analysis, ensuring data integrity through preprocessing and outlier detection.
- Developed predictive models and visualizations with ggplot2, applying t-tests and confidence intervals to derive insights for strategic decision-making.

Inventory Management System | Java, JavaFX, SQL, Data Visualization Feb 2024 - Mar 2024

- Developed inventory Management System using JavaFX, providing an intuitive UI for managing customer orders and product inventory supporting over 1,000 transactions daily.
- Implemented database integration with optimized SQL queries for efficient data retrieval and dynamic charts to visualize sales data and inventory trends.

Real-time DDOS attack detection using Apache Flink and Gradient boosting algorithm | Apache Flink, Gradient Boosting, AWS,

Firestore Aug 2022 - Apr 2023

- Developed a machine learning model using Apache Flink and Gradient Boosting to predict DDoS attacks, achieving a 95% detection rate.
- Simulated DDoS attacks on AWS, capturing over 10,000 requests in CSV format for analysis with 98% accuracy via a web app utilizing Firestore's real time database for instant updates.

SKILLS

- **Languages:** Python, SQL, Scala, R, Java, C, HTML, CSS
- **Data Analysis & Visualization:** MySQL, MongoDB, Power BI, Tableau, Google Analytics
- **Machine Learning & AI:** Pandas, NumPy, Matplotlib, Scikit-Learn, Seaborn, TensorFlow, PyTorch, OpenAI API
- **Tools & Frameworks:** Agile, Scrum, JIRA, ITIL, COBIT, SDLC, Lean Six Sigma, CI/CD Pipelines
- **Cloud & Platforms:** AWS (S3, EC2, ECR), Google cloud, Microsoft Azure, Apache Spark, Hadoop, Docker, GitHub
- **Certifications:** Google data analytics, Oracle Cloud Data Management 2023 Certified Foundations Associate, Foundations of project management by google.